Name_____ Date_____ Period_____

LESSON 5-1 · HOMEWORK PACKET · SHEET 1 OF 2

Area of Parallelograms

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a parallelogram by multiplying the base by the perpendicular height.

REMEMBER

- Cut a triangle off a parallelogram and slide it over to make a rectangle: **A = b × h**.
- The height *h* is **perpendicular** to the base. Do not use the slanted side.
- Area is in **square units** (cm², in², ft²).

WORKED EXAMPLE

Find the area: base 15 in, slanted side 8 in, height 6 in.

Step 1 Use the perpendicular height: h = 6 (skip the 8 in side)

Step 2 A = b × h = 15 × 6

Step 3 Multiply: 15 × 6 = 90

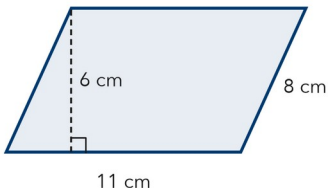
Answer **90 in²**

PRACTICE

show your work · label units

1. Find the area of the parallelogram.

Show your work.



Answer_____ cm²

2. Find the area of each parallelogram with base *b* and height *h*.

a. b = 12 ft, h = 7 ft _____ ft²

b. b = 9 m, h = 9 m _____ m²

c. b = 20 in, h = 4 in _____ in²

d. A = 63 m², h = 7 m; b = _____ m

3. **Parking Lot.** A lot has 10 parking spaces. Each space is a parallelogram with a base of 9 ft and a perpendicular height of 18 ft. What total area do the spaces cover?

Show your work.

Answer_____ ft²

4. **Mural.** A mural is painted on a parallelogram-shaped wall section. The area is 108 ft² and the base is 12 ft. What is the perpendicular height?

Answer_____ ft

5. **Find the Error.** A parallelogram has a base of 10 cm, a slanted side of 7 cm, and a height of 5 cm. Kiara says its area is 70 cm². Explain her mistake and find the correct area.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME Find a parallelogram shape at home or on the street (a tile, a sign, a parking space). Show someone where its base and height are.

Name_____ Date_____ Period_____

LESSON 5-1 · HOMEWORK PACKET · SHEET 2 OF 2

Area of Rhombuses

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a rhombus using the base and height or the two diagonals.

REMEMBER

- A rhombus is a parallelogram, so **A = b × h** works.
- With the **diagonals**: **A = (d1 × d2) ÷ 2**. Do not forget to divide by 2.

WORKED EXAMPLE

Find the area: diagonals 12 cm and 9 cm.

Step 1 Multiply the diagonals: 12 × 9 = 108

Step 2 Divide by 2: 108 ÷ 2 = 54

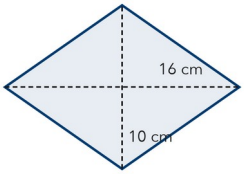
Answer **54 cm²**

PRACTICE

show your work · label units

1. Find the area of the rhombus.

Show your work.



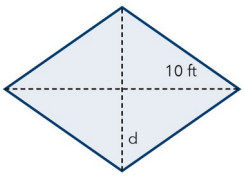
Answer_____ cm²

2. Find the area of each rhombus. Choose the formula that fits.

a. diagonals 14 m and 10 m _____ m² **b.** base 9 ft, height 7 ft _____ ft²

3. **Window.** A rhombus window has an area of 60 ft². One diagonal is 10 ft. Find the other diagonal *d*.

Answer_____ ft



4. **Stretch.** Rhombus A has diagonals of 12 ft and 8 ft. Rhombus B has a base of 8 ft and the same area as Rhombus A. What is the height of Rhombus B?

Answer_____ ft

5. **Find the Error.** Ravi says a rhombus with diagonals of 18 in and 8 in has an area of 144 in². Explain his mistake and find the correct area.

_____

_____

TEST PRACTICE

choose the best answer

6. A rhombus has sides of 7 m and a perpendicular height of 5 m. What is its area?

A 12 m² **B** 17.5 m² **C** 35 m² **D** 49 m²

7. A rhombus has diagonals of 20 in and 9 in. What is its area?

A 29 in² **B** 45 in² **C** 90 in² **D** 180 in²

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___